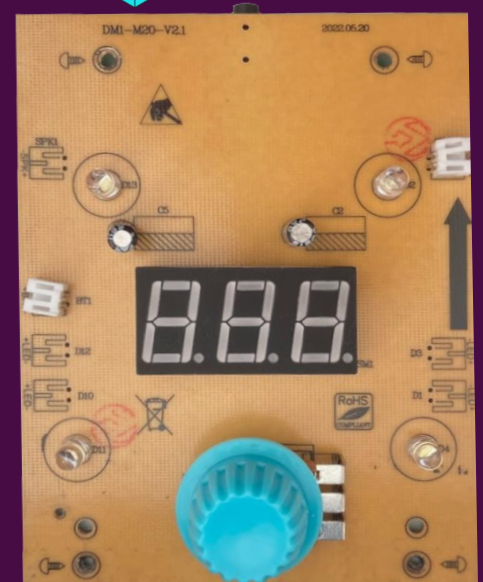
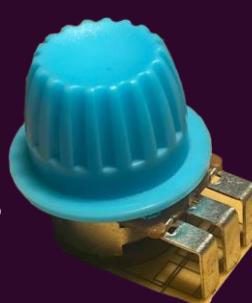




These are both very inexpensive components that contain all of the processing power in the product – highlighting the overall simplicity of the dance mat.



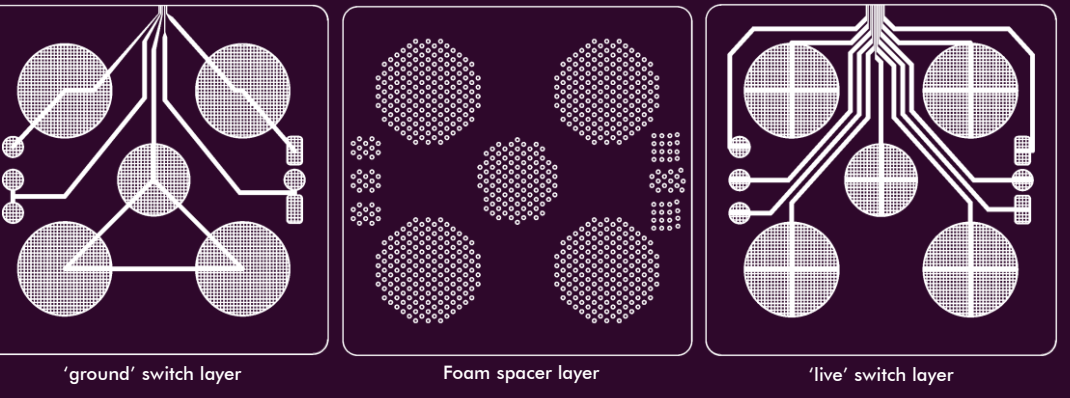
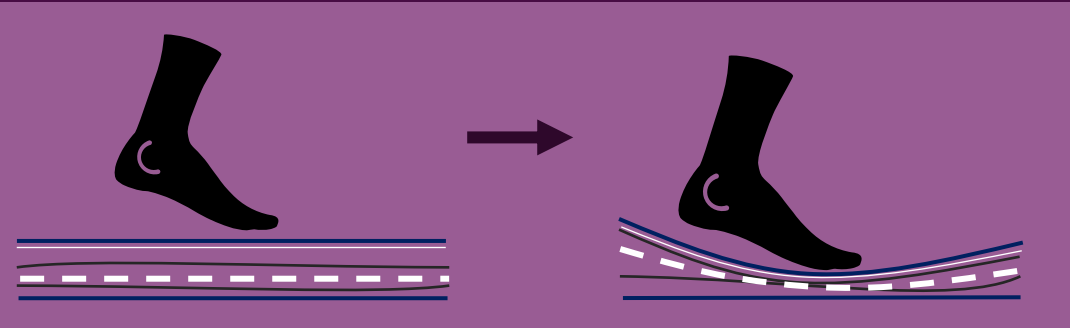
Component Role & Function

Deep Dive: Dance Mat Layers

Layers like an onion

Membrane switches

The mechanism for activating the switches inside the mat is to physically push down the upper plastic membrane until it contacts the lower membrane through the holes in the middle spacing foam – much like common keypads or computer keyboards! The foam spacer layer provides the mechanical resistance when the user steps on it. This resistance could be varied by altering the physical properties of the foam such as reducing hole density or increasing the thickness of the foam.

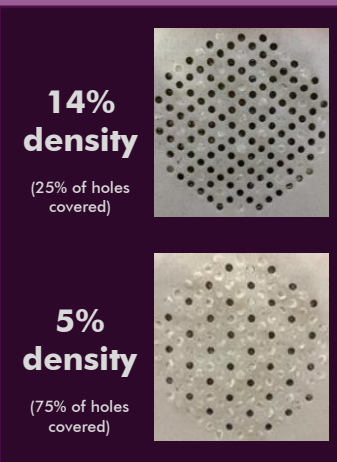
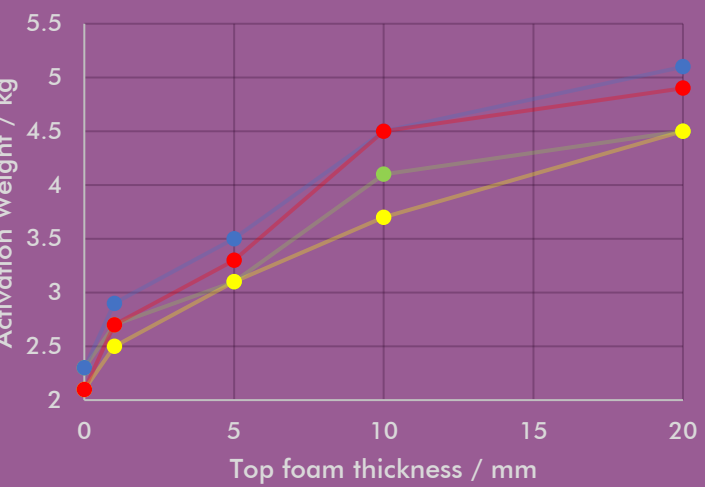


The ground and live switch membranes are transparent PET films with printed conductive graphite tracks. The schematics (above) show the circular areas where the switches can be pressed and the dense 5mm grid inside each circle to ensure every step is registered. This is a very simple and effective mechanism with minimal moving parts and no complex electronic components to detect weight or pressure on the switches. The switch membranes are also reliable and very cheap to mass produce although quite fragile and repeated folding causes creases which will eventually make the graphite chip and peel off the PET.

Foam layers

There are 2 foam layers in the mat: one on top of the first printed circuit and one between the printed circuits. The first provides protection for the flexible membranes and makes the mat softer to stand on. The second provides the spacing between the switch membranes and has holes where the switch layers contact (see schematic above). Compression of this layer provides the resistance to pressing the switches.

The effect of changes to the top and mid foam have been tested and quantified below. The hi-tec setup to test activation weight involved taping a hardback book to the top of and shoe and then loading it with household objects of known weights until the switch activated (see image - FYI bananas weigh approximately 400 g).



Conclusions

The top graph shows that increasing the thickness of the top foam layer distributes the user's weight more effectively – requiring more force to push through the thicker foam layer. However, testing with several different users showed this was not an issue as the activation weight only increases by ~2 kg for a very thick layer of foam which is not noticeable when jumping around! User feedback on the thicker foam layer was very positive: “more comfortable”, “bouncier!”, “more supportive”. Another important feature of the first foam layer is to add friction between the top PEVA layer and the PET membrane as these are 2 very smooth, slippery materials.

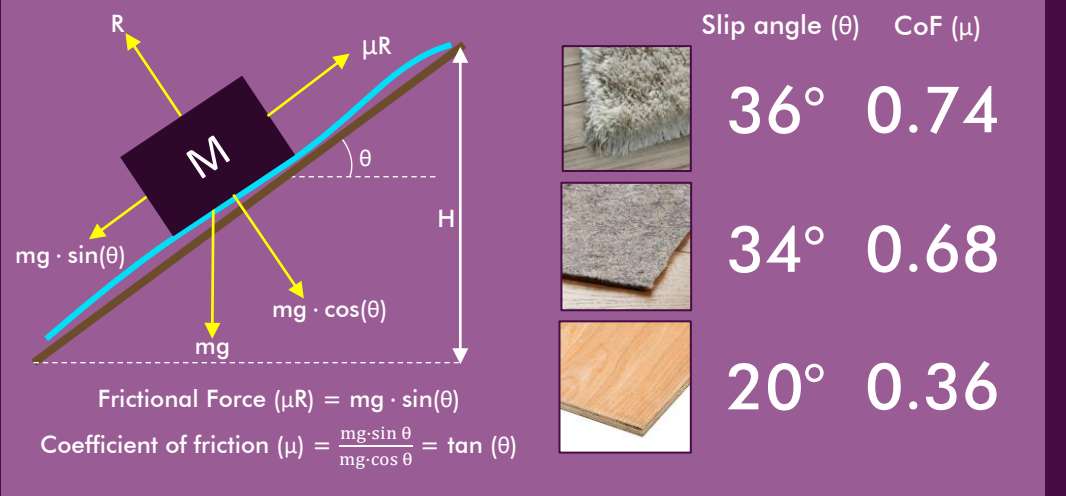
“More fun!” “Extra comfortable!” “Bouncier” “More supportive”

The second graph shows the effect of varying hole coverage in middle foam layer between the 2 switch membranes. 25%, 50% and then 75% of the holes were filled in. This had the effect of both increasing the amount of material between the membranes and reducing the surface area through which the membranes could contact each other through – both of which increased the activation weight. This did not noticeably impact the user feel, however, the 2 tests done here could be combined to create a thicker spacer foam layer with a higher hole density to simultaneously provide a bouncier and more supportive platform whilst also keeping the same activation weight.

How slippery is non-slip?

Coefficient of friction testing

The bottom of the mat is textured PEVA. It is advertised that this is to make it non-slipping when you are using it. Enthusiastic dancing with several users has confirmed that the effect is achieved successfully – but to what extent? Using an incline plane covered with different materials it is possible to quantify just how ‘non-slip’ the surface is by estimating the coefficient of friction (μ) between the mat and the floor.



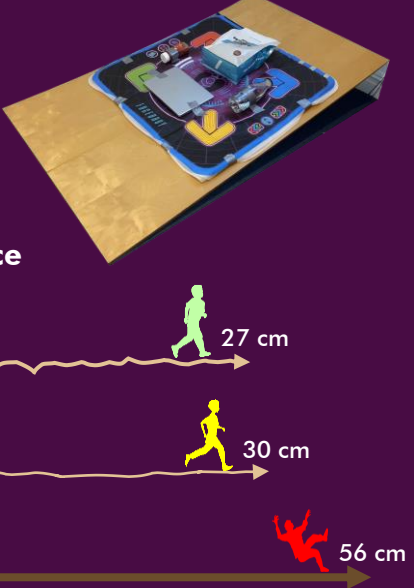
Test setup

Two dismantled wardrobe doors were angled against a wall to create the incline plane. 5 kg of household objects were taped to the surface of the mat which was then laid on the incline plane. In order to find μ (coefficient of friction), the incline angle (θ) must be increased until the point the mat starts to slip down the slope. Using some trigonometry (see above) an estimated coefficient of friction between the mat and the surface being tested can be found.

This test was carried out on 3 surfaces: **smooth wood**, **thin carpet** and **deep-pile thick carpet**. This data can be used for relevant performance-based calculations such as how far the mat will slip if a child jumps on it.

Data processing

Using the distance between the buttons (50cm) and assuming the fastest a button can be pressed is 0.25s the estimated speed is 2m/s. Along with the average mass of a 10-year-old (40kg - 392 N), the slippage of the mat can be calculated as shown below.

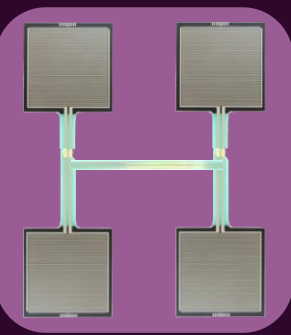


Conclusions

The quality of this data is questionable at best based on the makeshift scientific setup; however, it does successfully highlight how the performance of the non-slip base varies according to the surface the mat is used on. During normal use, the mat does not slip at all due to the user not transferring all their weight or momentum as fast as they can on every jump. Overall, although the mat is quite slippery on smooth wood the performance of the non-slip base is good enough for all normal use cases

Improvements

The dance mat is notably well optimised for simplicity, reliability and cost. The layers in the mat are simple, cheap to produce in bulk and the switch mechanism is extremely reliable. There is very little scope to change components in the product without negatively affecting the cost or complexity.



Adjustable force sensing

Popular in the dance mat hobby scene, using force sensitive resistors (right) is a more effective way to reliably control the switch activation points. These can be programmed to respond to certain weights – a setting or dial could be added to the product to give the user control of this function.

Furthermore, this would remove the need for 2 separate switch and foam layers as these resistors could be on one layer with 1 separate foam layer to provide cushioning. This also removes the foam switch mechanism which relies on compression of the middle foam layer to control activation force. Replacing the switches would represent quite a significant cost increase compared to the flexible membranes in place, but the ability to reprogram the mat along with a more reliable switch system may appeal to many users who are willing to pay a premium.

Foam layer alternatives

A layer of foam at the bottom of the mat, underneath both switch layers, would have the effect of adding cushioning to the mat but not affecting the activation pressure required. This was tested by adding a layer of 30mm thick EPE foam under one of the switches. The result was a very fun and bouncy experience! However, the thick, squishy foam would prevent the mat from being folded and transported/stored easily – one of the key selling points. Alternatively, a denser foam (such as the closed-cell PE sheet shown) would reduce the thickness required whilst retaining the soft platform.

Combining these two improvements (force sensitive resistors and denser foam layers) would be more effective. It would reduce the layers of the dance mat from 6 to 4 whilst increasing the bounciness of the mat for the same thickness of foam.

